SENTINEL

Overview · How to Use · Cheatsheet

TrinTech Digital Defense · v3.16.21 · 5332ca6

Continuous host monitoring and early warning for Caribbean SMBs. Authorized defensive use only — Detect · Alert · Report.

Generated 2026-10-30 22:08 AST · Pilot candidate — not a certified production release until release gates are evidenced.

1. What Sentinel is

Sentinel is TrinTech Digital Defense's continuous monitoring and early-warning engine for SMBs in Trinidad & Tobago and the Caribbean. It runs on a host (Linux preferred; Android/Termux for light pilots), watches for suspicious processes, persistence, file integrity changes, SSH brute-force activity, and canary-file access, and can alert via WhatsApp, email, webhook, or SIEM.

Deployment model: Install on **one always-on sensor** for a pilot. That host is protected deeply; other LAN devices are discovered and can be covered with optional remote agents. Sentinel is not a full SOC, EDR, or perimeter firewall.

Core capabilities

Capability	Default posture	Notes
File integrity (FIM)	Baseline then check	Menu 5 — run once after install
Persistence hunting	On in hunt loop	Cron, systemd, keys, LD_PRELOAD surfaces
Reverse-shell / behavioral	Detect + alert	Heuristics — not proof alone
Canary / deception files	Deploy on demand	Menu 4 — touch triggers alert
SSH brute-force	Detect; auto_block OFF	Opt-in host DROP via nft/iptables
Host firewall (nft/iptables)	Dry-run unless confirm	Root required for real DROP
Always-on service	Optional install	systemd or background hunt
Local dashboard	View-only by default	127.0.0.1 + token; PDF reports
Multi-host agents	Optional	Central listener + install bundles
Field evidence pack	On demand	Hashed zip for pilot handoff
Alerts (WA / email / SIEM)	Off until configured	Dedup, quiet hours, offline queue

2. How to use (pilot path)

Install

```
git clone https://github.com/trintechdigitaldefense/Sentinel.git
cd Sentinel && python3 setup.py
python3 sentinel.py autosubnet
python3 sentinel.py doctor
python3 sentinel.py
```

First-hour checklist

- **Menu 1** — START continuous protection (full stack)
- **Menu 5** — Create FIM / integrity baseline
- **Menu 4** — Deploy canaries if needed
- **Menu 13** — Always-on service install
- **Menu 16** — Doctor — fix FAILs / review WARNs
- **Menu 18** — Go-live readiness when ready
- Optional: Menu 22 Ops → PDFs, or python3 sentinel.py dashboard

Weekly / biweekly supervision

- Run doctor and review WARNs
- Generate client PDF (dashboard or client-report)
- Review critical alerts; ACK false positives
- Confirm canaries; re-deploy if needed
- Review SSH candidates; enable auto_block only if intended

Dashboard

Start with python3 sentinel.py dashboard. Open the printed URL (includes ?token=...). Client and technical PDF buttons stream a real PDF. Safe actions (doctor, ops, service status, visibility, multi-host, field evidence) work in view-only mode. Mutating actions need dashboard.allow_control: true.

SSH brute-force & firewall

Detection runs in the hunt loop; **auto_block is off** by default. Real DROP needs root + nftables or iptables. Prepare: sudo python3 sentinel.py firewall-prepare then sudo python3 sentinel.py nft-apply --confirm. Android/Termux is usually detect-and-record only.

Multi-host & field evidence

Hub: start central, multi-host prepare, enroll agents. Handoff pack: python3 sentinel.py field-evidence (SHA-256 zip under ~/.sentinel/data/). Hash is local integrity — not a full legal chain of custody.

3. Main menu cheatsheet

Run python3 sentinel.py with no arguments.

#	Action
1	START continuous protection (full stack)
2	Quick network scan (one-shot)
3	Hunt only (continuous) — Ctrl+C to stop
4	Deploy deception / canary traps
5	Integrity baseline (FIM + SUID)
6	Scan suspicious processes
7	Show full report
8	Detection tools →
9	System info
10	View alerts
11	Hardening audit
12	Status + ops metrics
13–15	Always-on service: install / status / stop
16	Doctor (health check)
17	Auto-detect scan subnets
18	Go-live readiness
19	Pilot validation workflow
20	Firewall / nftables tools →
21	SSH brute-force tools →
22	Ops (PDF, dashboard, cases) →
23	Fleet + agents (multi-host) →
24	Phase 4 advanced tools →
25	Schedule WhatsApp digest
0	Exit

Submenus (highlights)

20 Firewall: status, prepare backend, nft-apply, operator protect, defense pack, intel.
21 SSH: status, check now, expire blocks; whitelist/unblock via CLI.
22 Ops: client/technical PDF, dashboard, cases, offline queue, evidence, SIEM, suppressions.
23 Fleet: fleet scan, agent bundle, keys, canaries, central, multi-host status, field evidence.

4. CLI cheatsheet

Command	Purpose
python3 setup.py	Single install / wiring
python3 sentinel.py	Interactive menu
python3 sentinel.py doctor	Health checklist
python3 sentinel.py start	Scan + deception + one cycle
python3 sentinel.py hunt	Continuous hunt loop
python3 sentinel.py deception	Deploy canaries
python3 sentinel.py integrity	FIM baseline
python3 sentinel.py service-install	Always-on hunt
python3 sentinel.py service-status	Service + heartbeat
python3 sentinel.py dashboard	Local web UI
python3 sentinel.py client-report	Plain-language PDF
python3 sentinel.py pdf	Technical-style PDF
python3 sentinel.py ssh-blocks check	SSH failure scan
python3 sentinel.py firewall-prepare	Root + nft/iptables ready?
python3 sentinel.py nft-apply --confirm	Apply nft base ruleset
python3 sentinel.py multi-host status	Agent fleet status
python3 sentinel.py multi-host prepare	Agent install bundle
python3 sentinel.py field-evidence	Pilot evidence zip
python3 sentinel.py field-soak --confirm	Lab soak + sims
python3 sentinel.py go-live	Readiness report
python3 sentinel.py suppress-add TYPE pat	Mute false positive
python3 sentinel.py autosubnet	Safe local /24 only

5. Important paths & defaults

• Config: ~/.sentinel/config.json (mode 0600 when written by Sentinel)
• Data: ~/.sentinel/data/ — alerts, baselines, canaries, reports, evidence zips
• SSH auto_block: **false** by default
• Firewall: dry-run unless confirm / explicit apply
• Dashboard: loopback bind; view-only until allow_control
• Subnets: prefer single local /24 — avoid scanning 10.0.0.0/8

6. Positioning for TrinTech clients

Sell as continuous visibility and early warning on a managed sensor — not a replacement for backups, patching, or a full SOC. Ideal engagement: fixed-price pilot (install + baseline + always-on + weekly check-in), then expand to multi-host agents where needed. Field evidence packs support handoff and internal review; use precise language on hash integrity vs legal custody.

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